

Information Systems

Application Development for Information Systems

IDENTIFICATION

CODE : IF-3-DASI
ECTS : 2.0

HOURS

Lectures : 10.5 h
Seminars : 4.0 h
Laboratory : 32.0 h
Project : 0.0 h
Teacher-student
contact : 46.5 h
Personal work : 10.0 h
Total : 56.5 h

ASSESSMENT METHOD

Exam and projects

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. GRIPAY Yann
yann.gripay@insa-lyon.fr
MME TCHOUNIKINE Anne
anne.tchounikine@insa-lyon.fr

AIMS

The objective of this course is to study methods and tools for the development of service oriented applications for information systems (IS). We will study transactional multi-tier architectures and Web technologies for user interfaces.

The target skills in this course are:

- 1) Apply methods for analysis and design of IS:
 - Identify business objects, business services, and associated GUI
 - Design and implement data models
- 2) Design and implement applications for IS:
 - Write IS specifications documents
 - Implement an application from IS specification documents

CONTENT

- Bases of analysis and design of applications for IS
- Specifications of the 3 layers: interface-business-data
- Persistence and object-relational mapping technologies (ORM)
- GUI Design Principles
- Architectures MVC (Model-View-Controller)
- Web Technologies for GUI

Technologies: Java, JPA (persistence), Servlets, JavaScript, AJAX

PRE-REQUISITE

UML Modeling
Databases
Algorithmic and Object-oriented Programming
Java development